

Spacing Out: Retrieval Practice, Sleep and Exam Performance in an Introductory Programming Course

Dara Sok and Mei Lin Tan · Learning Sciences Demo Lab

Sample document for the AI Summary demo. The study, participants and data are fictional.

Abstract

Students in introductory programming courses often prepare for exams by re-reading notes in the days before the test. We compared this massed review with spaced retrieval practice, in which students answered short low-stakes quizzes spread across the term. In a fictional cohort of 212 first-year students, the spaced retrieval group scored 78% on the final exam, compared with 64% for the massed review group, a difference of 14 percentage points (Cohen's $d = 0.62$). Students who reported at least seven hours of sleep before the exam retained more of what they had practised, and the benefit of spacing was largest for students who also slept well. We argue that course designers can build retrieval practice and sleep-aware deadlines directly into the structure of a course.

1 Introduction

Learning to program asks students to hold many new ideas in mind at once: variables, control flow, functions and the habits of debugging. Many students respond by cramming, spending long sessions re-reading slides shortly before an exam. Re-reading feels productive because the material looks familiar, but familiarity is not the same as being able to recall and apply an idea.

Two findings from memory research suggest a better approach. Retrieval practice is the act of recalling information from memory, for example by answering a quiz question without looking at notes, and it strengthens later recall more than re-reading. The spacing effect describes the observation that practice distributed over several days produces more durable learning than the same amount of practice massed into one session. Sleep also matters, because memories are consolidated during sleep after learning.

This study asks whether combining spaced retrieval practice with attention to sleep improves exam performance in a first-year programming course, and whether the two factors interact.

2 Method

The fictional cohort included 212 first-year students enrolled in an introductory Python course over a twelve-week term. Students were randomly assigned by tutorial group to one of two study conditions.

In the spaced retrieval condition, students completed a ten-question online quiz every Tuesday and Friday. Each quiz mixed questions from the current week with questions from earlier weeks, and students received feedback immediately after answering.

In the massed review condition, students received the same questions as a printed study guide with worked answers, and were encouraged to review it before the exam. Both groups had access to identical lectures, labs and teaching assistants.

Students kept a short sleep diary during the final week, recording the hours they slept each night. The final exam contained 40 questions on reading, tracing and writing code, and was marked by teaching assistants who did not know which condition each student belonged to.

3 Results

Students in the spaced retrieval group averaged 78% on the final exam, while students in the massed review group averaged 64%. The difference of 14 percentage points corresponds to a medium-to-large effect size ($d = 0.62$).

The benefit was clearest on code-writing questions, where the spaced retrieval group scored 21 percentage points higher. On simple definition questions the gap was only 6 points, which suggests that retrieval practice mainly helped students apply ideas rather than recognise them.

Sleep was associated with better performance in both groups. Students who slept at least seven hours on the night before the exam scored 9 points higher on average than students who slept less than five hours.

The two factors interacted: spaced retrieval students who also slept well scored highest, averaging 84%. In contrast, 31% of students in the massed review group reported studying past midnight before the exam, and this group had the lowest average score at 57%.

4 Discussion

The results support a simple conclusion: students learn programming more durably when they repeatedly retrieve ideas over time, and when they protect their sleep before assessments. Frequent low-stakes quizzes turn every week into a small opportunity to practise recall, and mixing older questions into new quizzes provides spacing without extra planning from students.

Course design can make these habits the default. Instructors can schedule short weekly quizzes, move assignment deadlines away from late-night submission times, and avoid scheduling major deadlines on the night before an exam.

5 Limitations

Because the data are fictional and the sleep measure is self-reported, the numbers should be read as an illustration rather than evidence. A real study would also need to account for prior programming experience, attendance and motivation, which can differ between tutorial groups.

6 Conclusion

Spaced retrieval practice produced higher exam scores than massed review, and the advantage was largest for students who slept at least seven hours before the exam. Building retrieval practice and sleep-aware deadlines into a course is a low-cost way to help novice programmers learn more.

References

- Cepeda, N. J., Pashler, H., Vul, E., Wixted, J. T., and Rohrer, D. (2006). Distributed practice in verbal recall tasks: A review and quantitative synthesis. *Psychological Bulletin*, 132(3), 354–380.
- Roediger, H. L., and Karpicke, J. D. (2006). Test-enhanced learning: Taking memory tests improves long-term retention. *Psychological Science*, 17(3), 249–255.